Question

$\mathbf{M}$ can directly react with carbon monoxide under certain conditions to obtain carbonyl compound $\mathbf{A}$. $\mathbf{A}$ reacts with sodium metal in liquid ammonia at low temperature to obtain salt $\mathbf{B}$ containing a tetrahedral mononuclear anion. It is known that the mass fraction of $\mathbf{M}$ in $\mathbf{A}$ is 34.47%. $\mathbf{B}$ can react with carbon tetrachloride in a 3:1 ratio to generate $\mathbf{C}$ with C_{3v} symmetry. Adding the chloride of $\mathbf{M}$ to a concentrated ammonium chloride-ammonia solution, and adding hydrogen peroxide dropwise, can prepare cation $\mathbf{D}$.

Determine the correctness of the following conclusions related to the above content, and select the option corresponding to all the correct conclusions.

Conclusion 1: The mass fraction of $\mathbf{M}$ in $\mathbf{C}$ is 37.11%.

Conclusion 2: The reaction of a metal element with an atomic number 1 greater than that of $\mathbf{M}$ with carbon monoxide to obtain a carbonyl compound can occur under milder conditions.

Conclusion 3: The mass-to-charge ratio of $\mathbf{D}$ is 54.04.

Conclusion 4: There is only one type of ligand in $\mathbf{D}$.

Conclusion 5: The reaction of the iodide of $\mathbf{M}$, copper metal, and carbon monoxide can yield $\mathbf{A}$, while generating a salt with a mass fraction of 20.03% for one of the metals.

A. All other options are incorrect

B. Conclusion 1, Conclusion 2, Conclusion 3

C. Conclusion 1, Conclusion 2, Conclusion 4

D. Conclusion 1, Conclusion 2, Conclusion 5

E. Conclusion 1, Conclusion 3, Conclusion 4

F. Conclusion 1, Conclusion 3, Conclusion 5

G. Conclusion 1, Conclusion 4, Conclusion 5

H. Conclusion 2, Conclusion 3, Conclusion 4

I. Conclusion 2, Conclusion 3, Conclusion 5

J. Conclusion 2, Conclusion 4, Conclusion 5

K. Conclusion 3, Conclusion 4, Conclusion 5

Answer

Correct Answer: B

Detailed Explanation

The general formula of $\mathbf{A}$ should be $\mathrm{M(CO)}_x$, where x can be 4, 4.5, 5, etc. Through trials, it is found that when $x = 4$, $M(\mathbf{M}) = [4 \times 28.01/(1 - 0.3447)] \times 0.3447 = 58.94(\mathrm{g/mol})$. The molar mass is closest to that of cobalt, followed by nickel.

CHECKPOINT 1 PTS

$M(\mathbf{M}) = [4 \times 28.01/(1 - 0.3447)] \times 0.3447 = 58.94(\mathrm{g/mol})$

If $\mathbf{M}$ is cobalt, then $\mathbf{A}$ is $Co_2(CO)_8$, which can be reduced to $[Co(CO)_4]^-$ by metallic sodium in liquid ammonia, which meets the condition of "tetrahedral mononuclear anion" in the question.

If $\mathbf{M}$ is nickel, then $\mathbf{A}$ is $Ni(CO)_4$, which can be reduced to $[NiH(CO)_3]_2$, $[Ni_4(CO)_9]^{2-}$ and other species by metallic sodium in liquid ammonia, all of which do not meet the condition of "tetrahedral mononuclear anion" in the question.

Therefore, $\mathbf{M}$ is cobalt, $\mathbf{A}$ is $Co_2(CO)_8$, and $\mathbf{B}$ is $Na[Co(CO)_4]$.

CHECKPOINT 2 PTS

$\mathbf{M}$ is cobalt, $\mathbf{A}$ is $Co_2(CO)_8$, and $\mathbf{B}$ is $Na[Co(CO)_4]$.

$Na[Co(CO)_4]$ can react with carbon tetrachloride in a 3:1 ratio to generate $ClC[Co(CO)_3]_3$, so **C** is $ClC[Co(CO)_3]_3$.

CHECKPOINT 1 PTS

C is $ClC[Co(CO)_3]_3$

Calculate the mass fraction of Co in it:

$$w(Co) = \frac{3 \times 58.93}{35.45 + 12.01 + 3 \times (58.93 + 3 \times (12.01 + 16.00))} \times 100\% = 37.11\%$$

CHECKPOINT 1 PTS

$w(Co) = \frac{3\times58.93}{35.45+12.01+3\times(58.93+3\times(12.01+16.00))} \times 100\% = 37.11\%$

Therefore, conclusion 1 is correct.

The direct reaction of cobalt with carbon monoxide generally takes place at 100 atm and 200 degrees Celsius. The metal element with an atomic number 1 greater than Co is Ni, and the typical conditions for the reaction of Ni with carbon monoxide to obtain a carbonyl compound are 1 atm and 50 degrees Celsius, and the reaction conditions are milder. Therefore, conclusion 2 is correct.

CHECKPOINT 1 PTS

The direct reaction of cobalt with carbon monoxide generally takes place at 100 atm and 200 degrees Celsius. The metal element with an atomic number 1 greater than Co is Ni, and the typical conditions for the reaction of Ni with carbon monoxide to obtain a carbonyl compound are 1 atm and 50 degrees Celsius, and the reaction conditions are milder.

By adding cobalt chloride to a concentrated ammonium chloride-ammonia solution and dropping in hydrogen peroxide, the cation $[Co(NH_3)_5(H_2O)]^{3+}$ can be prepared. Calculate its mass-to-charge ratio:

CHECKPOINT 1 PTS

the cation $\mathbf{D}$ is $[Co(NH_3)_5(H_2O)]^{3+}$

$$m/z = (58.93 + 5 \times (14.01 + 3 \times 1.008) + (16.00 + 2 \times 1.008)) \div 3 = 54.04$$

CHECKPOINT 1 PTS

$m/z = (58.93 + 5 \times (14.01 + 3 \times 1.008) + (16.00 + 2 \times 1.008)) \div 3 = 54.04$

Therefore, conclusion 3 is correct.

$[Co(NH_3)_5(H_2O)]^{3+}$ contains two types of ligands, ammonia molecules and water molecules, so conclusion 4 is incorrect.

CHECKPOINT 1 PTS

$[Co(NH_3)_5(H_2O)]^{3+}$ contains two types of ligands, ammonia molecules and water molecules

The equation for the reaction of Co iodide, elemental copper, and carbon monoxide is

$$2CoI_2 + 4Cu + 8CO \rightarrow Co_2(CO)_8 + 4CuI$$

The only salt produced is CuI, and the only metal element in it is copper. The mass fraction of copper is

$$w(Cu) = \frac{63.55}{63.55 + 126.9} \times 100\% = 33.37\%$$

which is not equal to 20.03%, so conclusion 5 is incorrect.

In summary, choose option B.